

Assignment 1: Polynomial Regression

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Phase 1 (var1): turbine power score AND Phase 2 (var2): thermal anomaly score

GitHub repository: https://github.com/harsh-kumar-005/ML_Assignment_1

1. Summary

	var1	var2
Selected model	Lasso, degree 5, all 6 features	Ridge, degree 11, all 3 features
Penalty (alpha)	0.007079	1.500000
Polynomial terms (non-zero)	461 (149)	363 (363)
Selection CV MSE	0.340560	0.229650
Repeated 5x5 CV MSE $\pm$ std	0.342 $\pm$ 0.029	0.230 $\pm$ 0.019
Repeated 5x5 CV R ²	0.9673	0.9945
80/20 hold-out MSE / R ²	0.305 / 0.9718	0.235 / 0.9960
Training MSE / R ²	0.230 / 0.9784	0.158 / 0.9964

All model-selection metrics use training rows and labels only. Test labels are unavailable, and test inputs are not used for degree, regularisation, model-family, or feature-selection decisions. Test inputs are used only after the final model is fixed to generate predictions.

2. Data exploration

Each training file has 1000 rows and each test file has 1000 rows. There are no missing values. A small number of duplicate rows occur in the test inputs; these inputs are never used for model selection. Features lie in $[-1, 1]$, with many values exactly at the bounds.

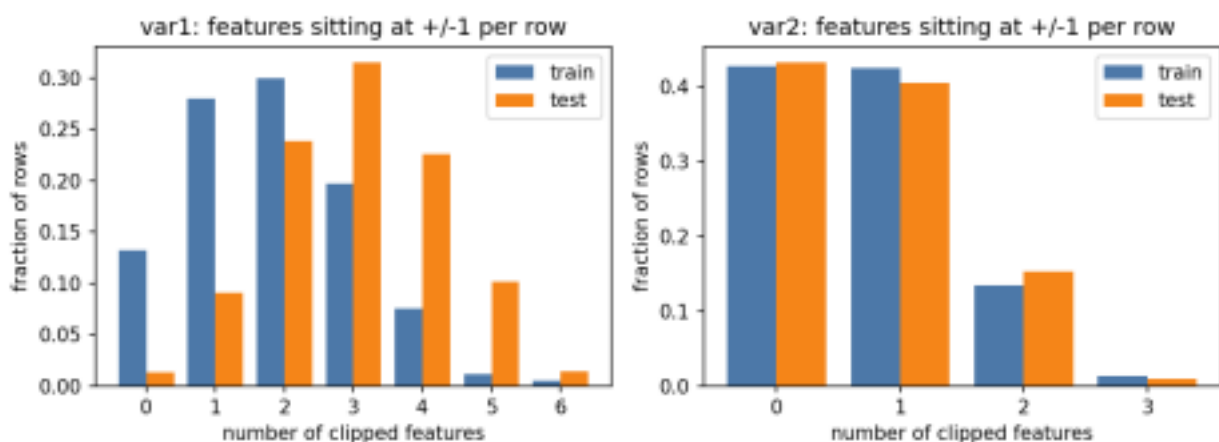


Figure 1. Boundary/clipping diagnostics. The train/test comparison is descriptive only and does not enter model selection.

3. Method

Pipeline: PolynomialFeatures(degree) → StandardScaler → estimator, with OLS, Ridge (L2), Lasso (L1), or ElasticNet considered. Hyper-parameters were selected by K-fold cross-validation using only training rows and labels. Ridge is a pure L2 estimator, so l1_ratio is not applicable.

Selection: var1 uses the minimum training-only CV MSE over alpha and l1_ratio at degree 5. Var2 uses a training-only Ridge search over degrees 8–13 and alpha, selecting the degree/alpha pair with the minimum mean CV MSE.

4. Results

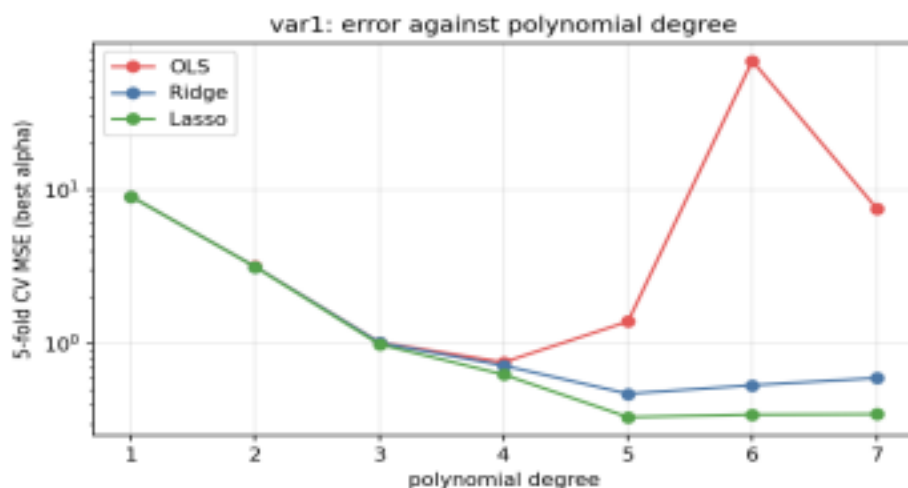


Figure 2. Var1 training-only cross-validated MSE against polynomial degree.

4.1 var1: turbine power score (6 features)

The table reports the best training-only CV MSE available within the corresponding alpha grid at each degree. Degree 5 is the selected degree for the final Lasso model.

Degree	OLS	Ridge	Lasso
1	9.0262	9.0183	9.0151
2	3.1545	3.1519	3.1377
3	1.0176	1.0120	0.9868
4	0.7545	0.7172	0.6273
5	1.3884	0.4714	0.3320
6	68.5149	0.5350	0.3447

Error falls steeply through degree 5 and rises sharply at degree 6 for OLS. The final degree-5 Lasso has 461 polynomial terms, with 149 non-zero coefficients. The alpha/l1_ratio search selects pure L1 (l1_ratio=1.0), so the final Var1 model is Lasso rather than an interior ElasticNet mixture.

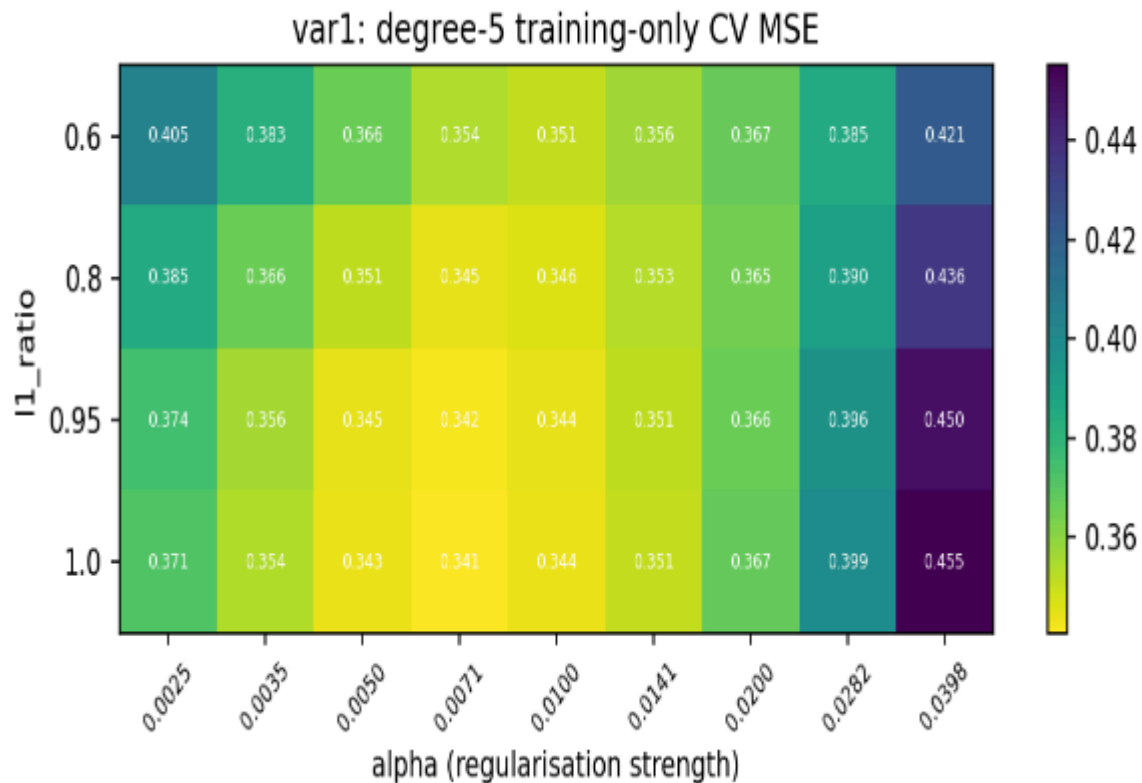


Figure 3. Var1 training-only CV MSE over alpha and l1_ratio at degree 5.

Final Var1 selection CV MSE = 0.340560; repeated 5×5 diagnostic CV MSE = 0.341697 ± 0.028891 .

Boundary stress test (training rows only): fit on 909 rows with at most 3 clipped features and evaluate on 91 rows with 4 or more; MSE=0.366, $R^2=0.988$. This is diagnostic only and uses no test information.

4.2 var2: thermal anomaly score (3 features)

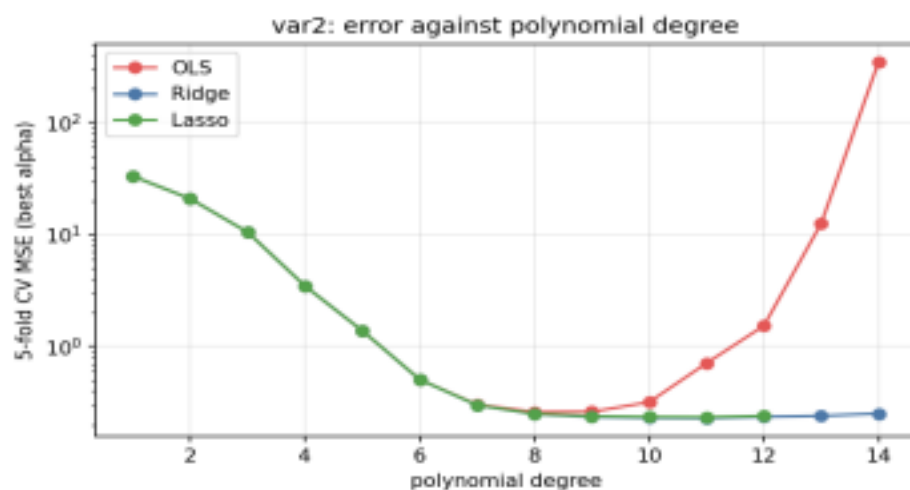


Figure 4. Var2 training-only cross-validated MSE against polynomial degree.

Initial degree sweep (degrees 1–7): the values below are the best training-only CV MSE within each model family's alpha grid. This table is exploratory; the final Var2 choice comes from the refined Ridge search below.

Degree	OLS	Ridge	Lasso
1	33.0977	33.0951	33.0977
2	20.8389	20.8352	20.8389
3	10.4848	10.4819	10.4848
4	3.4683	3.4682	3.4679
5	1.3822	1.3822	1.3819
6	0.5073	0.5066	0.5074
7	0.3037	0.2997	0.3004

Refined Ridge search (degrees 8–13): for each degree, the row below reports the minimum mean training-only CV MSE over the refined Ridge alpha grid.

Degree	Best Ridge alpha	Mean CV MSE	CV Std.	Decision
8	0.172331	0.248433	0.023422	candidate
9	0.533670	0.235396	0.019354	candidate
10	0.939133	0.230480	0.018376	candidate
11	1.500000	0.229650	0.014939	SELECTED
12	2.908281	0.236050	0.016913	candidate
13	2.908281	0.240013	0.014618	candidate

Final Var2 selection: degree 11 with Ridge alpha=1.5. Its mean training-only CV MSE is 0.229650, the minimum in the refined grid. Ridge is pure L2, so l1_ratio is not applicable.

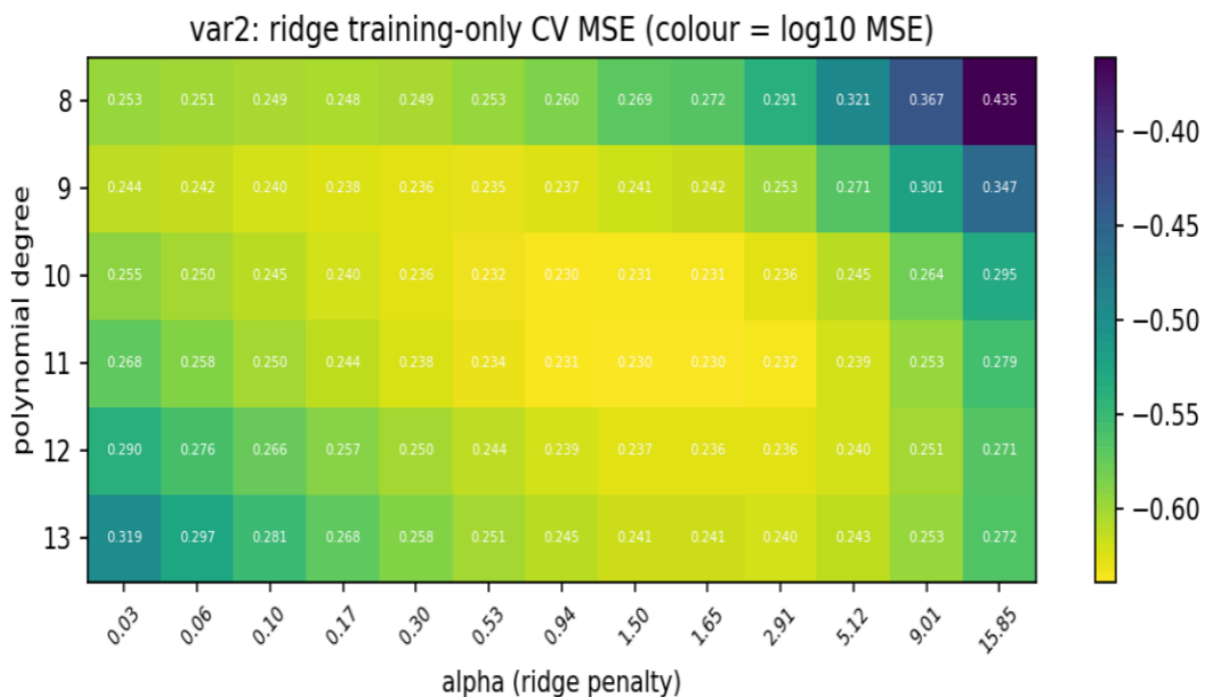


Figure 5. Var2 Ridge training-only CV MSE by degree and alpha.

Final Var2 model: degree 11, alpha=1.5, 363 polynomial terms, all 363 coefficients non-zero. Repeated 5×5 diagnostic CV MSE = 0.230094 ± 0.019121 ; hold-out MSE = 0.234906.

5. Feature ablation

Each feature was omitted in turn and the selected model was re-evaluated using the same training-only CV setup. The full-feature model is the baseline.

Problem	Feature omitted	CV MSE
var1	none	0.3409
var1	x1	3.1597
var1	x2	2.5715
var1	x3	3.5771
var1	x4	1.3801
var1	x5	4.5514
var1	x6	4.4420
var2	none	0.2282
var2	x1	23.0649
var2	x2	37.4583
var2	x3	17.2318

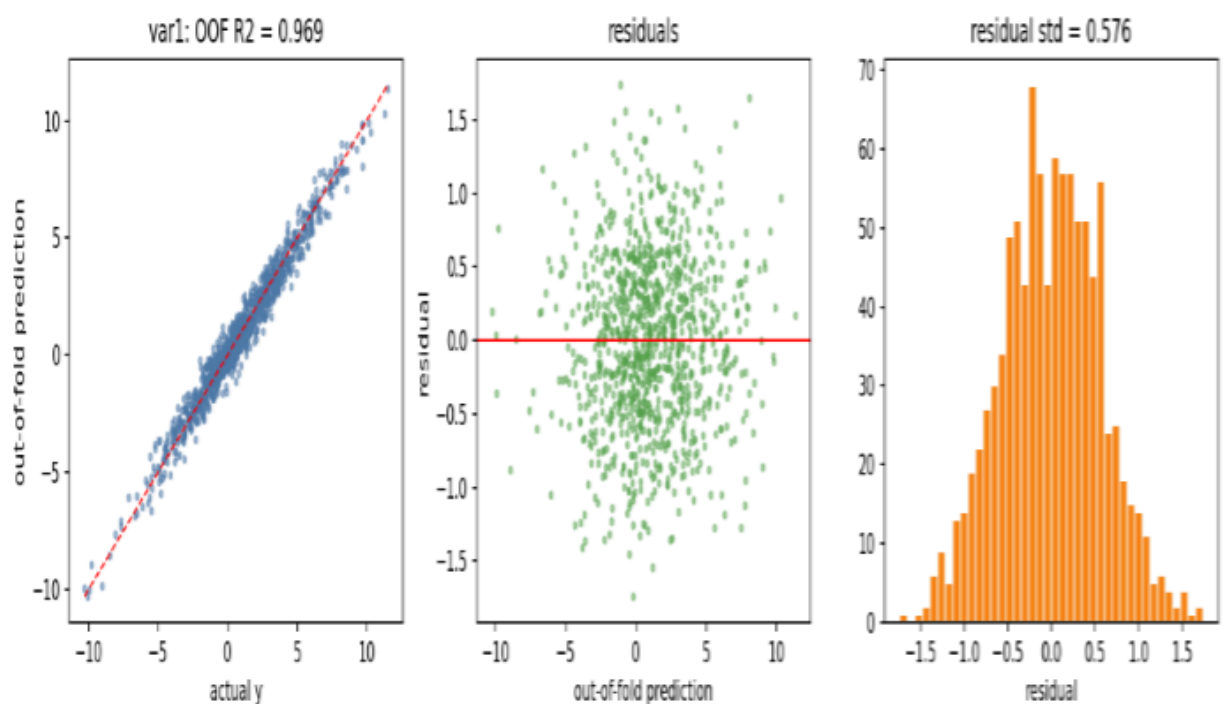


Figure 6. Var1 out-of-fold diagnostics.

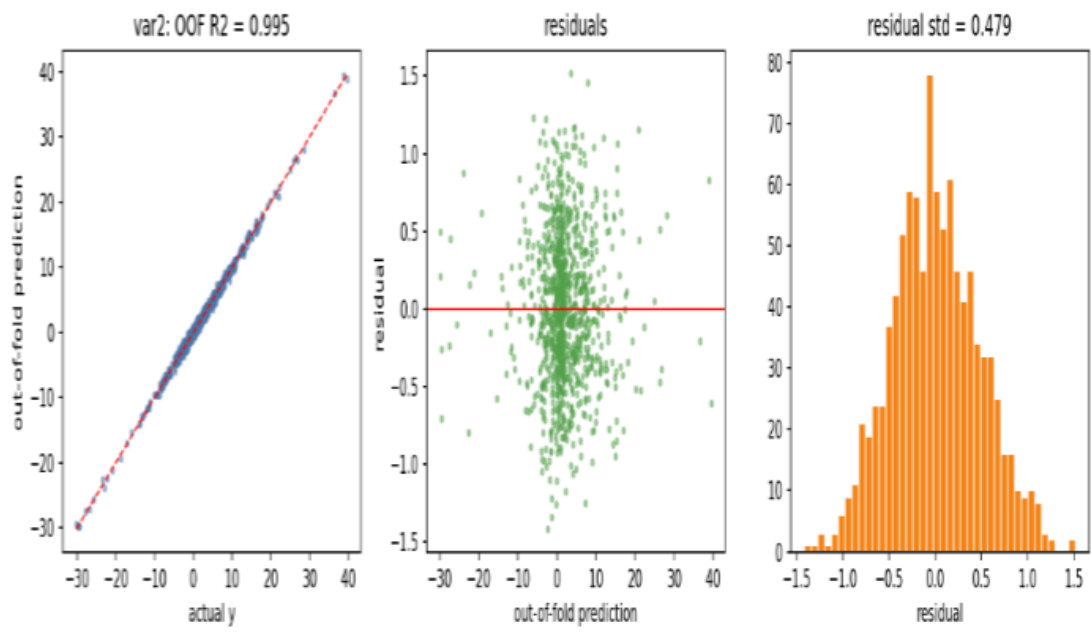


Figure 7. Var2 out-of-fold diagnostics.